Hypothesis Testing: The Foundations

Code

STAT 4714

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0.1 Goals for Today

- Understand the **logic of hypothesis testing**
- Explicitly define the **null hypothesis (H_0) and alternative (H_a)**
- Discuss ECE examples where hypothesis testing matters
- Briefly see the traditional formula-based approach
- Focus on the simulation-based view of testing

0.2 Quick Recap

- Sampling distributions describe what's **typical under repeated sampling**
- CIs and SE measure **precision**, not decisions
- Today: how to use sampling distributions to **make decisions**

1.1 Some Definitions

- **Hypothesis test**: compare our observed statistic to what we would expect under a null hypothesis
- **Null hypothesis (H_0)**: default claim; differences are due to chance/random variation, not a real effect
- **Alternative hypothesis (H_a)**: competing claim; there is a real difference or effect

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1.2 Ex: Voltage Stability

A circuit board is designed for $5.0V$. We measure sample voltages and see variation: $4.2V, 4.1V, 4.8V, 5.1V$. Let μ denote the true average voltage.

Hypotheses

- H_0: $\mu = 5.0V$
- H_a: $\mu \neq 5.0V$

1.3 Ex: Packet Loss

A network router should have $\leq 1\%$ packet loss. We test whether observed error rate is higher. Let p denote the true error rate.

Hypotheses

- $H_0: p = 0.01$
- $H_a: p > 0.01$

1.4 Ex: Material Strength

We compare two PCB designs A and B and want to know if material strength differs. Let μ_A and μ_B denote the true strength of designs A and B, respectively.

Hypotheses

- $H_0: \mu_A = \mu_B$
- $H_a: \mu_A \neq \mu_B$

1.5 Overview: Traditional Approach

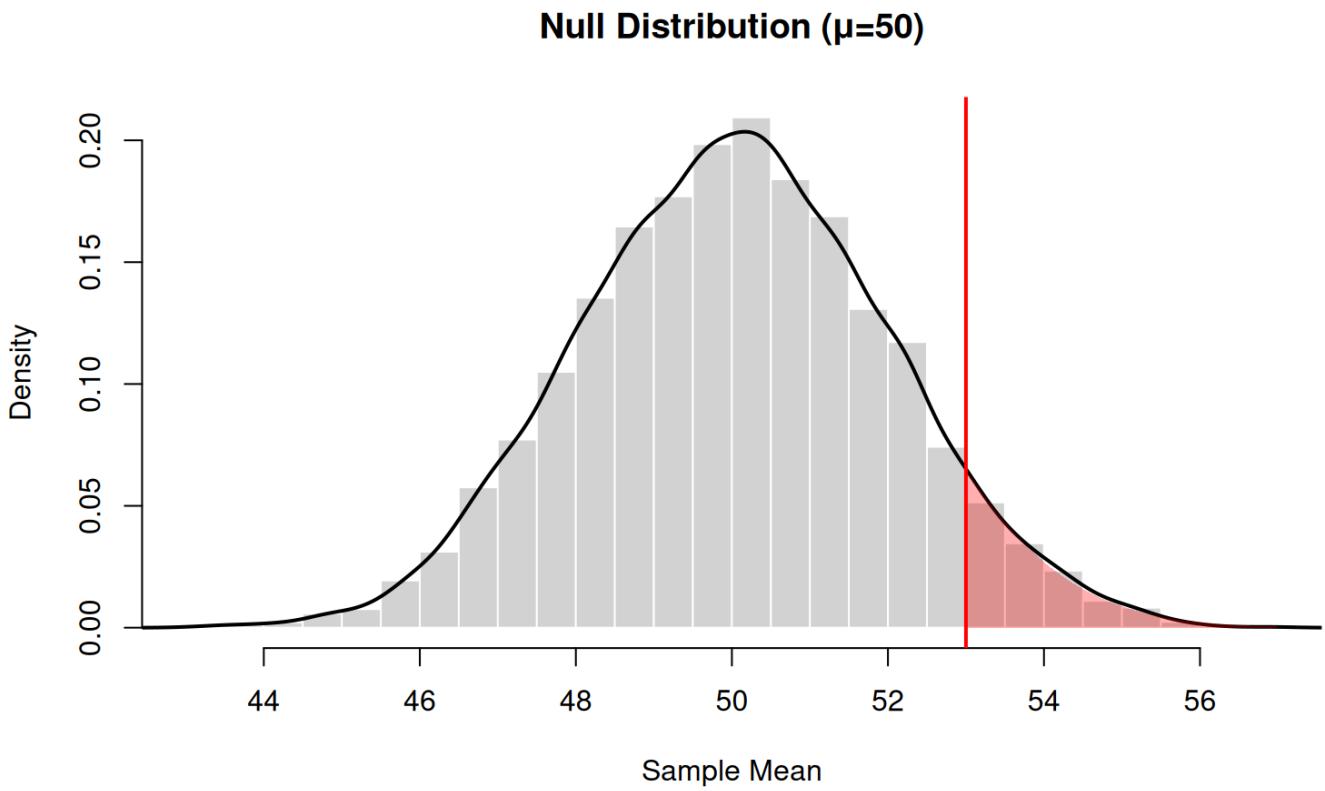
- Define H_0 and H_a
- Choose a test statistic (mean, difference, proportion, etc.)
- Compute the p-value from a known distribution (t, Z, etc.)
- Finally, make a decision to "reject" or "fail to reject" H_0 based on α

1.6 p-values

The p-**value** is the probability of observing a statistic under the null H_0 as extreme or more extreme than our observed statistic.

- A small p-value means that our observed statistic would be rare if H_0 were true; it's evidence that H_0 is probably wrong.
  - *Note:* The p-value does **not** tell us the probability that H_0 is true.

1.7 Visualizing p-values



The p-value is the shaded tail area beyond the observed sample mean under the null distribution.

1.8 Using the p-value

Pick a significance level α based on the desired confidence (e.g., $\alpha = 0.05$).

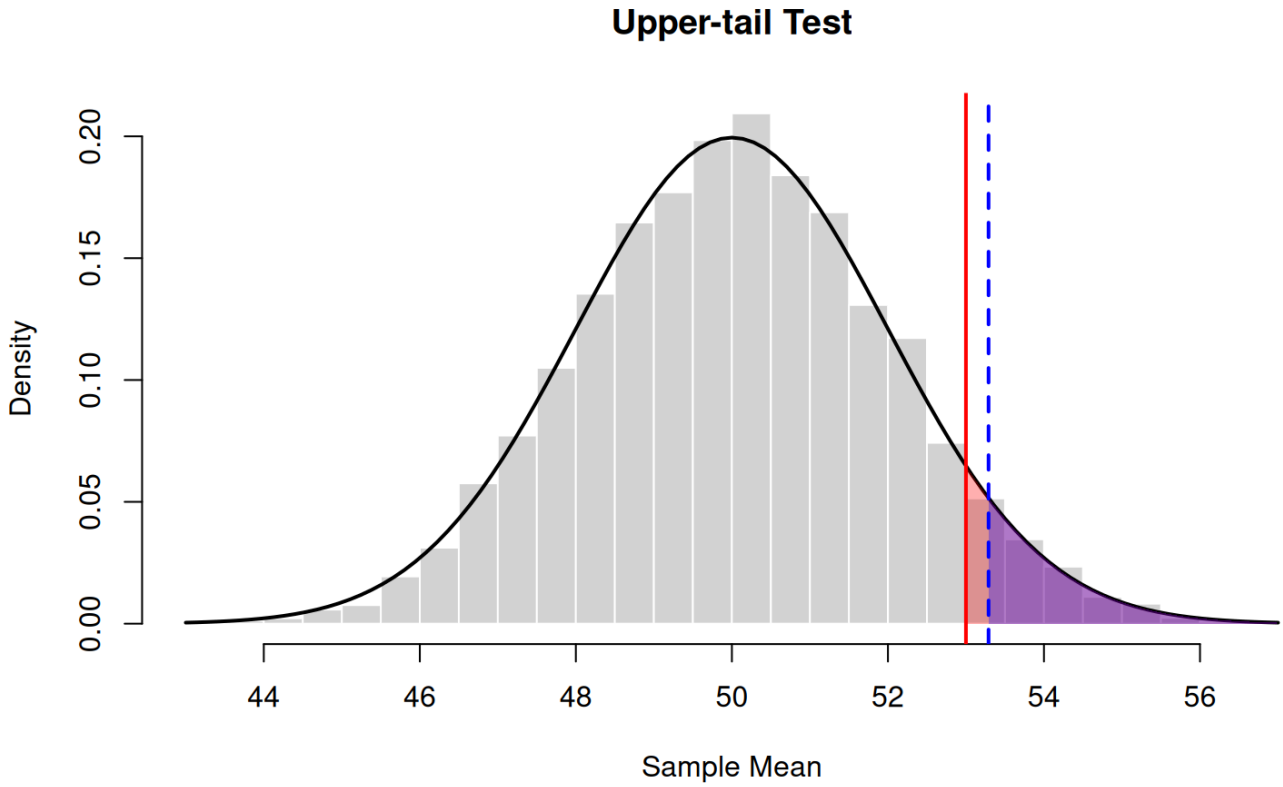
- If $p \leq \alpha$, reject H_0; there is evidence against the null
- If $p > \alpha$, fail to reject H_0; the data is consistent with the null
  - *IMPORTANT*: Rejecting H_0 is not proving that H_a is true, and failing to reject H_0 is not proving H_0 is true!
- The p-value is generally defined in terms of "extremeness." What is "extreme" depends on the research question…

1.9 Test Type: Upper-tail

$H_0: \mu = 50 \qquad H_a: \mu > 50$

"Extremeness" concerns the right (upper-tail)

$\alpha = 0.05$



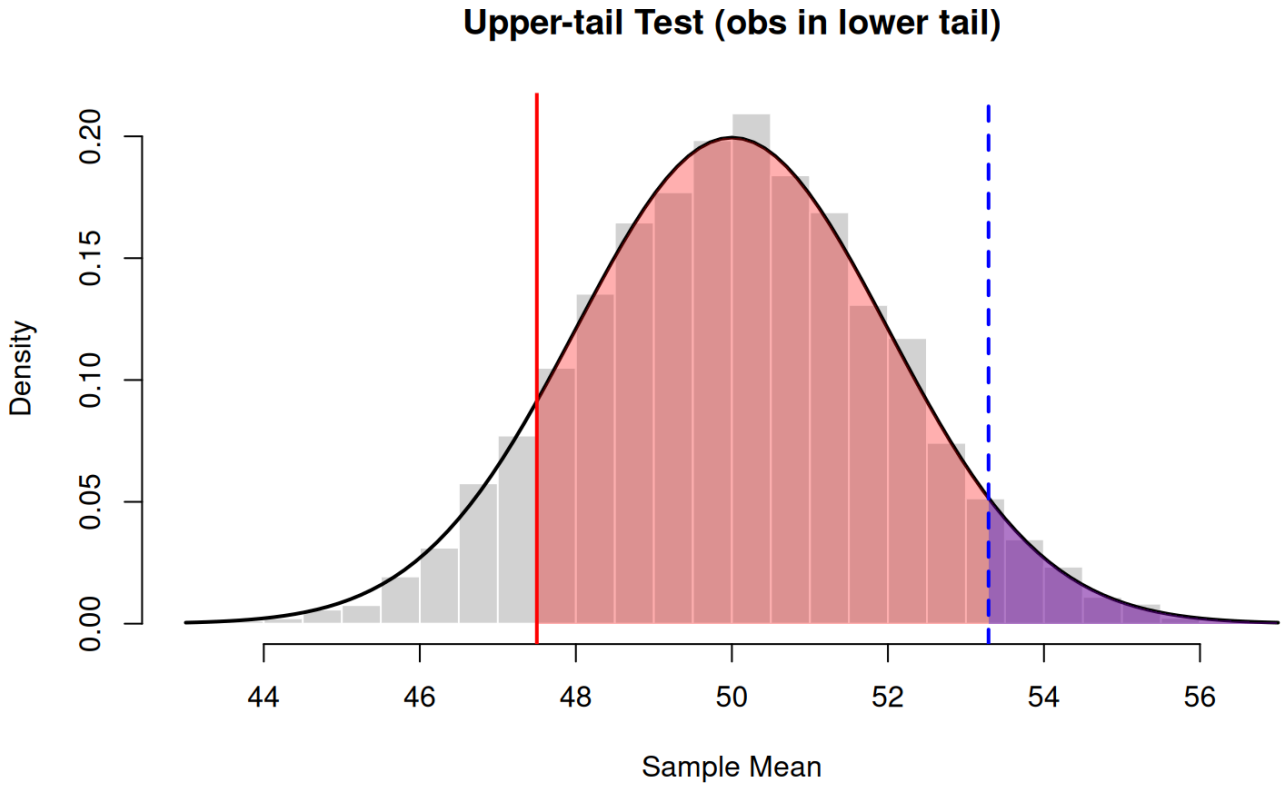
Upper-tail test: observed mean in red, p-value (red area), α region (blue area).

1.10 Test Type: Upper-tail

$H_0: \mu = 50 \qquad H_a: \mu > 50$

"Extremeness" concerns the right (upper-tail)

$\alpha = 0.05$



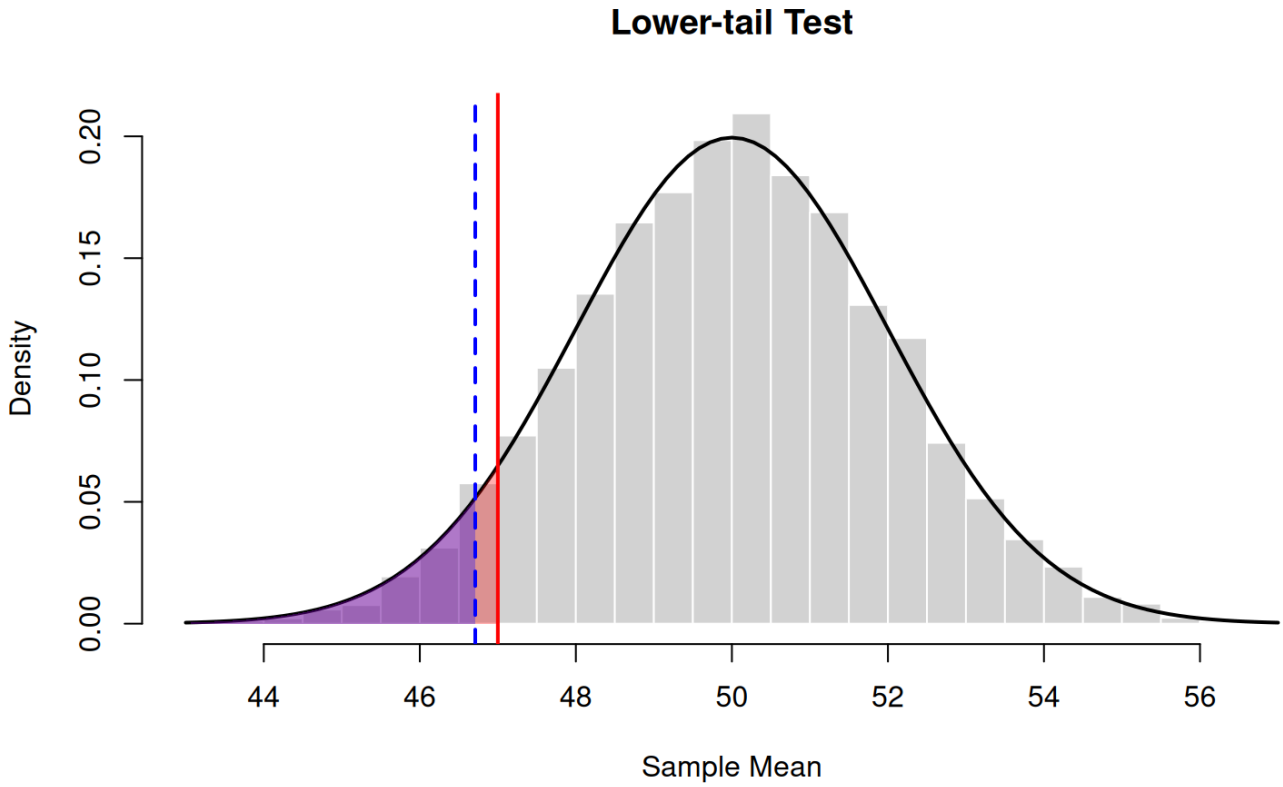
Upper-tail test: observed mean (red line) is in the lower tail. The p-value (red shaded area) is large because the observation is not in the direction of H_a.

1.11 Test Type: Lower-tail

$H_0: \mu = 50 \qquad H_a: \mu < 50$

"Extremeness" concerns the left (lower-tail)

$\alpha = 0.05$



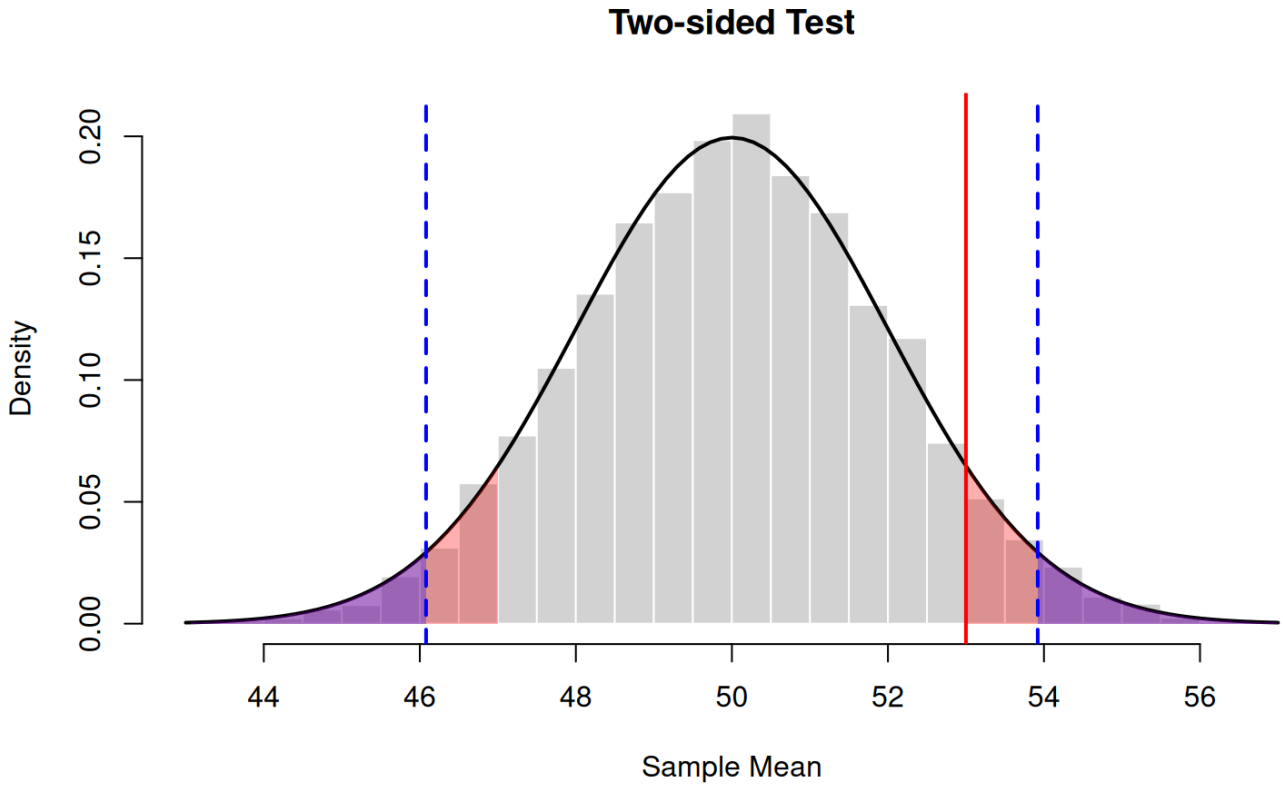
Lower-tail test: p-value (red area) and α region (blue area).

1.12 Test Type: Two-sided

$H_0: \mu = 50 \qquad H_a: \mu \neq 50$

"Extremeness" concerns the left and the right (both tails)

$\alpha = 0.05$



Two-sided test: p-value in both tails (red), α region in both tails (blue).

1.13 Sample mean (σ known): z-test

If we somehow know σ (rare in practice), the standardized mean is normally distributed.

- $H_0: \ \mu = \mu_0 \qquad H_a: \ \mu \neq \mu_0$
- Test statistic:

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

- p-value (two-sided): $p = 2\left(1 - F_Z(|z|)\right)$, where $F_Z(z)$ is the CDF of the standard normal distribution evaluated at z

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1.14 Sample mean (σ unknown): t-test

Usually σ is unknown, and we estimate it with s. Then, the standardized mean follows a **t** distribution.

- $H_0: \ \mu = \mu_0 \qquad H_a: \ \mu \neq \mu_0$
- Test statistic:

$$t \;=\; \frac{\bar{X} - \mu_0}{s/\sqrt{n}}, \quad \mathrm{df} = n - 1$$

- p-value (two-sided): $p = 2\left(1 - F_t(|t|;\ \mathrm{df})\right)$, where $F_T(x)$ is the CDF of the t distribution with df degrees of freedom evaluated at x

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1.15 Sample proportion: z-test (big n)

For pass/fail outcomes, the sample proportion is approximately normally distributed with large n.

- $H_0:\ p = p_0 \qquad H_a:\ p \neq p_0$
- Test statistic:

$$z \;=\; \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

- p-value (two-sided): $p = 2\left(1 - F_Z(|z|)\right)$, where $F_Z(z)$ is the CDF of the standard normal distribution evaluated at z

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1.16 Traditional Approach: Formulas ad Infinitum

- Many **case distinctions** (known vs unknown σ, one- vs two-sided, small n, normality assumptions, etc.)
- Lots of **reference distributions/multipliers** to remember
- Fortunately, we can build the **same logic** with **simulation under** H_0
  - This requires fewer formulas, is more intuitive, and yields the same decision

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2.1 The Main Idea

- Instead of memorizing formulas, **simulate under** H_0
  - Generate many samples (assuming H_0 is true) to build the null sampling distribution of the test statistic
- Compare our observed statistic to the null sampling distribution

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2.2 Comparing Approaches

1. Define H_0 and H_a
2. Choose a test statistic (mean, difference, proportion, etc.)
3. Compute the p-value from a known distribution (t, Z, etc.). Alternatively, compute the p-value from the sampling distribution under H_0
4. Use the p-value to make a decision to “reject” or “fail to reject” H_0 based on α

2.3 Ex: Circuit Board Voltage (#1)

Suppose a manufacturer claims their circuit boards run at an average of $5.0V$. We test a sample of 25 boards and observe a sample mean of $5.3V$. Assume that prior testing indicates the population standard deviation is $\sigma = 1.0V$ and that voltage is normally distributed. Is there evidence that the boards are running above $5.0V$ on average?

Our hypothesis:

- H_0: $\mu = 5.0V$
- H_a: $\mu > 5.0V$

2.4 Ex: Circuit Board Voltage (#2)

Suppose a manufacturer claims their circuit boards run at an average of $5.0V$. We test a sample of 25 boards and observe a sample mean of $5.3V$. Assume that prior testing indicates the population standard deviation is $\sigma = 1.0V$ and that voltage is normally distributed. Is there evidence that the boards are running above $5.0V$?

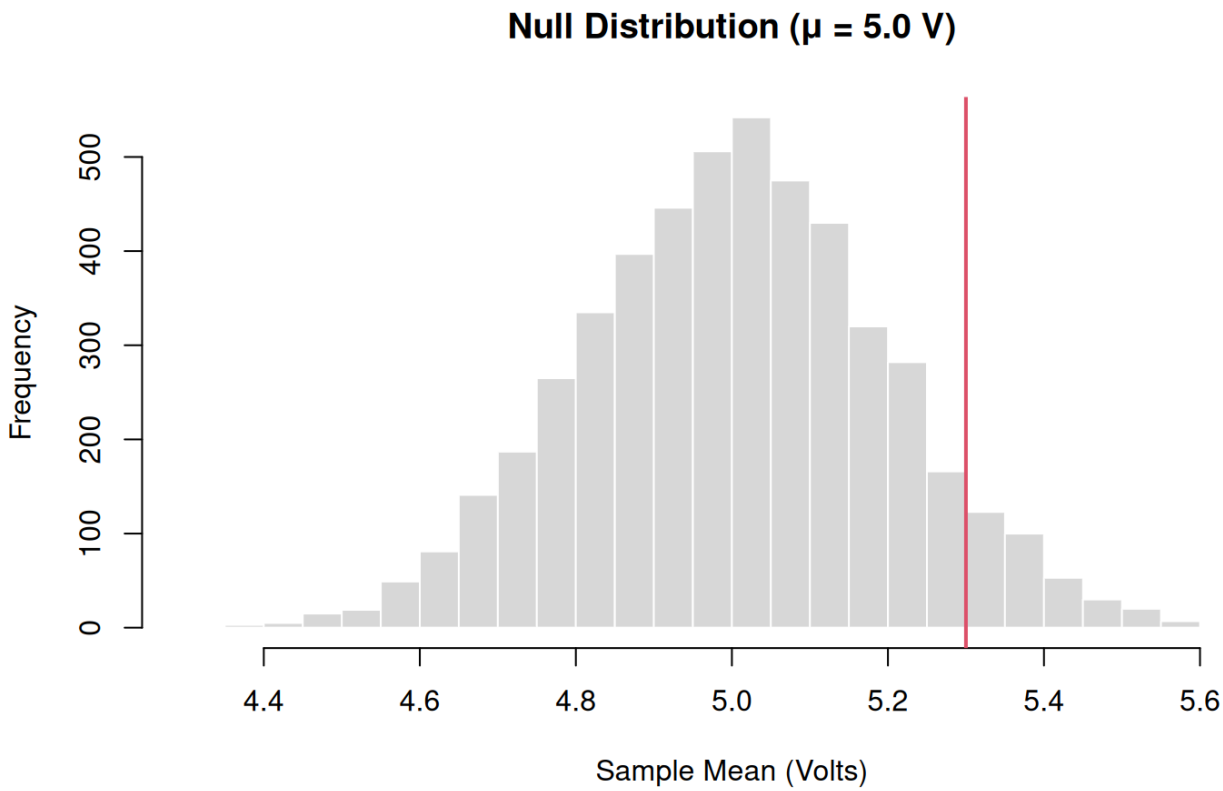
Test Statistic:

- The test statistic was chosen for us: we observe the sample mean $\bar{X} = 5.3V$.

2.5 Ex: Circuit Board Voltage (#3b)

Now, we compute the p-value from the sampling distribution under H_0: $\mu = 5.0V$ with $\sigma = 1.0V$ and $n = 25$. First, we get the sampling distribution:

▶ Code



Null distribution of sample means under H_0: μ=5.0 volts.

2.6 Ex: Circuit Board Voltage (#3b/#4)

Then, we compute the p-value from the sampling distribution. Recall that the p-value is the probability of obtaining a statistic (the sample mean) at least as extreme as what we observed **if H_0 were true**. So, we can simply count the proportion of simulated means greater than or equal to $\bar{X}_{obs} = 5.3$ volts:

▼ Code

```
p_val <- mean(null_means >= xbar_obs)
p_val
```

```
[1] 0.067
```

- With 95% confidence ($\alpha = 0.05$), we "fail to reject" H_0. With 90% confidence ($\alpha = 0.10$), we "reject" H_0.

2.7 Ex: Circuit Board Voltage (Recap)

- We generated competing hypotheses (H_0 vs H_a) based on the research question.
- Then, we generated a sampling distribution under H_0.
- Next, we compared the observed sample mean with the sampling distribution assuming H_0 was true.

- Finally, we saw that the observed statistic $\bar{X}_{obs}$ was in the right-tail, but it wasn't extreme enough to justify rejecting H_0.
    - **Main point**: Simulate assuming H_0, then test if our observation agrees with H_0.

2.8 Ex: Circuit Board Voltage (Trad. Solve)

To make the same decision using the traditional approach, we would need to compute the standardized test statistic

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}},$$

then find the associated p-value with its reference distribution (i.e., $Z \sim \mathrm{N}(0, 1)$). Finally, we compare the p-value to α and make our decision.

2.9 Ex: Packet Latency (#1)

Packet latencies are often right-skewed due to queueing and modeled with the log-normal distribution. A specification requires that the 95th percentile of latency must be less than 15ms. Suppose we observe 27 packet transmissions, and the 95th percentile of latency was 15.3ms. Assume that log-standard deviation is $\sigma = 0.6$ and log-mean is $\mu = 1.7211$. We want to know if the 95th percentile of latency is less than 15ms.

Our hypothesis:

- H_0: $q_{0.95} = 15$ms
- H_a: $q_{0.95} < 15$ms

2.10 Ex: Packet Latency (#2)

Packet latencies are often right-skewed due to queueing and modeled with the log-normal distribution. A specification requires that the 95th percentile of latency must be less than 15ms. Suppose we observe 27 packet transmissions, and the 95th percentile of latency was 15.3ms. Assume that log-standard deviation is $\sigma = 0.6$ and log-mean is $\mu = 1.7211$. We want to know if the 95th percentile of latency is less than 15ms.

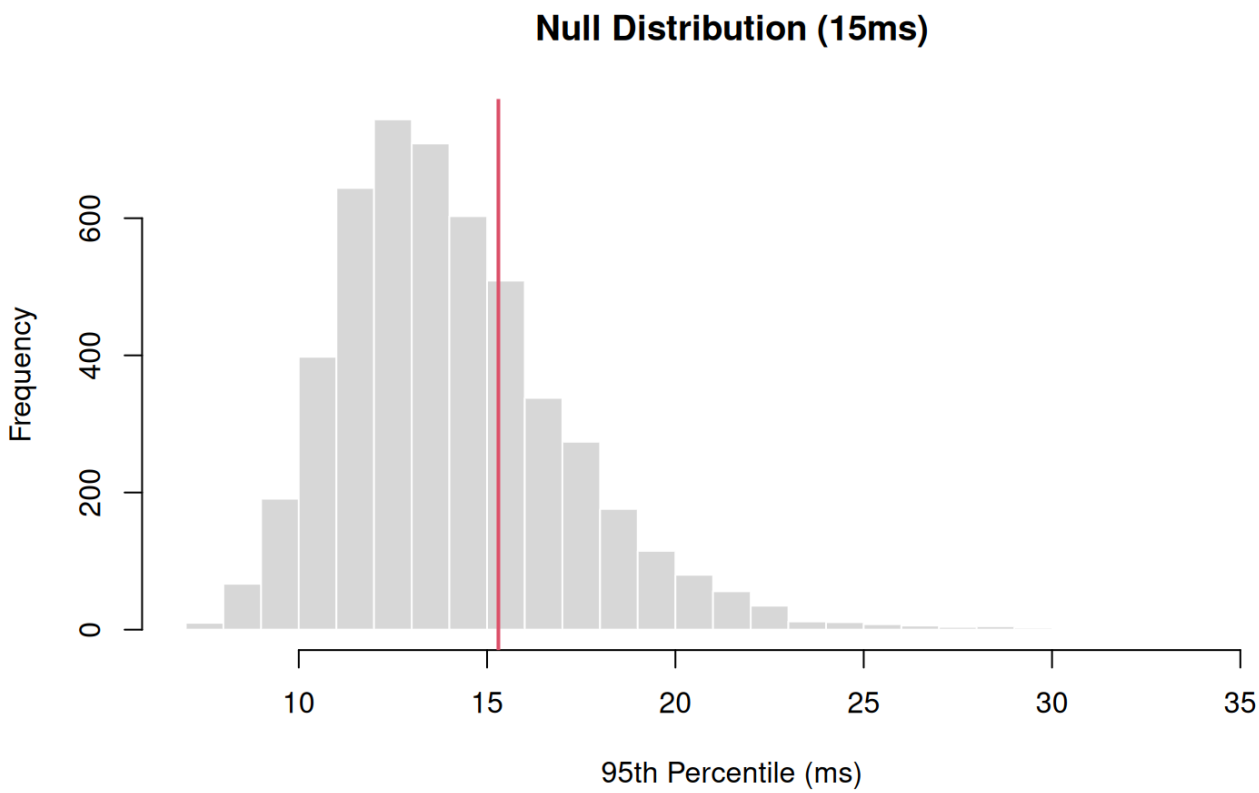
Test statistic:

- The test statistic was chosen for us: we observe the 95th percentile of latency as $q^*_{0.95} = 15.3$ms.

2.11 Ex: Packet Latency (#3b)

Now, we get the sampling distribution under H_0: $q_{0.95} = 15$ms with $\mu = 1.7211$, $\sigma = 0.6$, and $n = 27$:

▶ Code



Null distribution of 95th percentile under H_0: $q_{0.95} = 15$ms.

2.12 Ex: Packet Latency (#3b/#4)

Now, compute the p-value from the sampling distribution:

▼ Code

```
mean(null_percentiles <= percentile_obs) # less than because it's a lower-tailed test
```

```
[1] 0.7082
```

Our p-value is pretty large! Clearly, our observed value of 15.3 does not indicate that the 95th percentile is less than 15ms. Therefore, we "fail to reject" H_0.

- This does not indicate H_0 is true. It just means that H_a is less likely to be true than H_0.

2.13 Ex: Packet Latency (Trad. Solve)

- If $X \sim \text{Lognormal}(\mu, \sigma^2)$, then $\log X \sim N(\mu, \sigma^2)$.

- The population 95th percentile is

$$q_{0.95} = \exp(\mu + \sigma z_{0.95}),$$

  where $z_{0.95}$ is the 95th standard normal quantile.
- Under H_0, we fix $q_{0.95} = 15$ ms and assume sample size $n = 27$.
- **Key idea:** we want to know the distribution of the *sample 95th percentile* when drawing size $n = 27$ from this log-normal.

2.14 Ex: Packet Latency (Trad. Solve)

- For large n, the sample quantile $\hat{q}_p$ is approximately normal:

$$\hat{q}_p \approx \mathrm{N}\Big(q_p,\ \frac{p(1-p)}{n f(q_p)^2}\Big),$$

  where $f(q_p)$ is the PDF at the true quantile.
- Plugging in $p = 0.95$, $n = 27$, and $f(q_{0.95})$ from the log-normal PDF gives an **approximate standard error**.

2.15 Ex: Packet Latency (Trad. Solve)

- Test statistic (Z-score):

$$Z \ = \ \frac{\hat{q}_{0.95} - q_{0.95}}{\mathrm{SE}(\hat{q}_{0.95})}.$$

- For a **left-tailed test** $H_a : q_{0.95} < 15$ms:

$$p\text{-value} \ = \ P(Z \leq z_{\text{obs}}).$$

2.16 Ex: Packet Latency (Recap)

- The traditional approach requires memorizing/deriving four formulas. The simulation approach does not require any special formulas; you just need to simulate and compute the statistic correctly (1 line of code!).
- The simulation approach is more visual (at least to me).
- How we solved this packet latency provide is *very* similar to how we solved the circuit board voltage problem.
- Sampling distributions allow us to translate a hypothesis about the population into a testable question about the sample. Making decisions once we have the appropriate sampling distribution is simple.

3.1 Summary

- Hypothesis testing compares observed statistics to a **null distribution**

- Null hypothesis = default position; alternative = competing claim
- Traditional approach uses formulas; simulation offers intuition and flexibility
- Next: we'll extend hypothesis testing to **two-sample and proportion problems**

3.2 Breakout Work

1. Why do we need a **null distribution** at all?
2. What does it mean if our observed statistic is *not* in the tails?
3. Why might a simulation-based test be preferable to memorizing formulas?
4. Give one ECE example where a hypothesis test would guide a decision.